

# XPO: High Density Pluggable Data Center Optics

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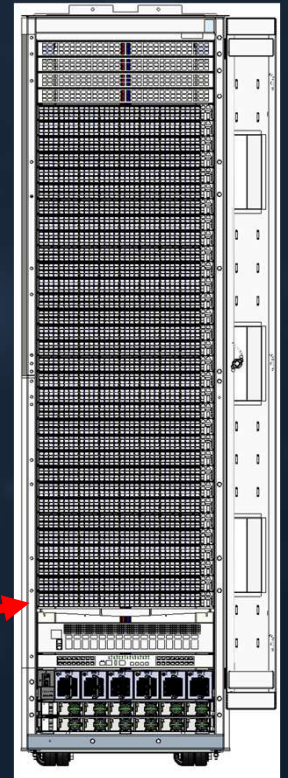
# AI Needs a Higher Density Form Factor

As More GPUs Operate as a Single System, Interconnect Bandwidth Must Scale with Compute.

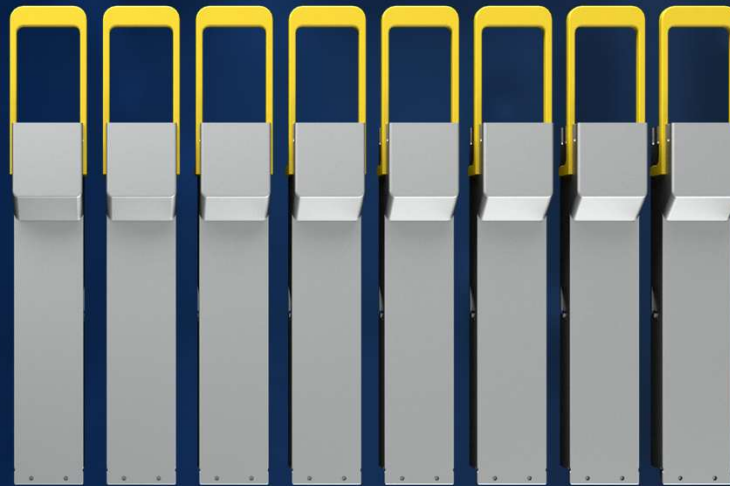
Single-chip 204.8T Switch with 128 pcs of 1600G-OSFP requires 4U Chassis so one switching Rack can provide only 1.6Pbps of bandwidth .

With 12.8T XPO, 204.8T fits into 10U so XPO based rack can provide 6.5Pbps Bandwidth per ORv3 Liquid Cooled Rack.

With 25.6T XPO, 409.6T fits into 10U so XPO based rack can provide 13.1Pbps Bandwidth per ORv3 Liquid Cooled Rack..



# XPO: A liquid cooled 12.8T Pluggable



One XPO replaces 8 OSFP

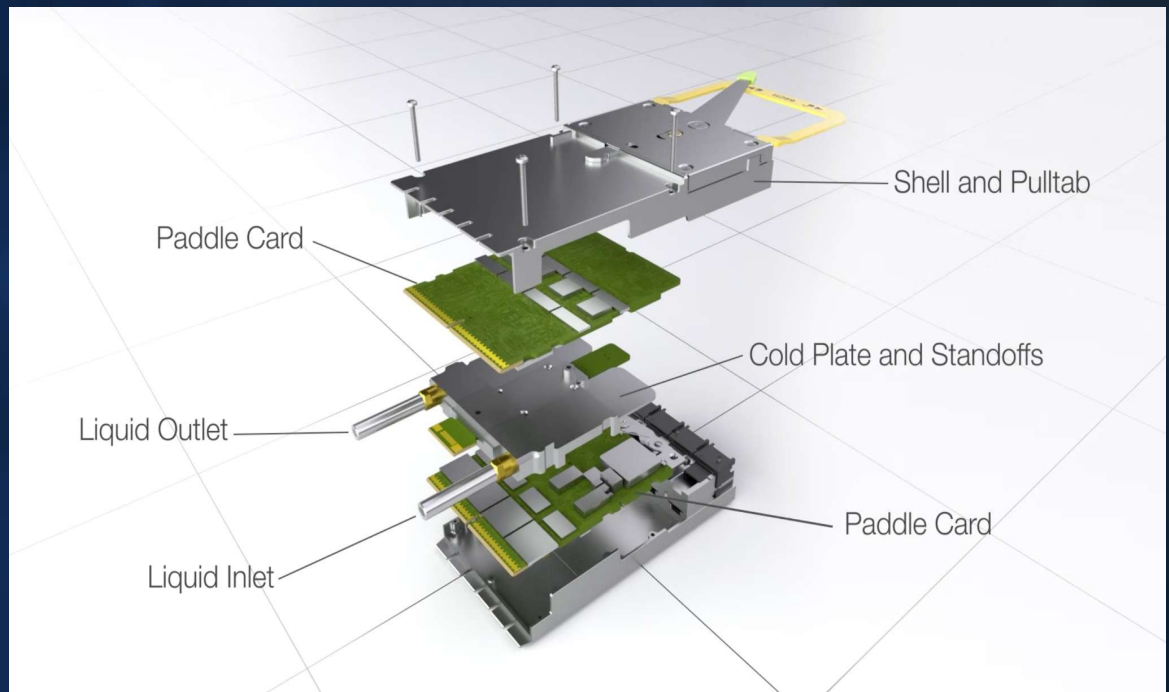
XPO Leverages the Proven OSFP Ecosystem for Faster Time-to-Market and Manufacturing Scale

# XPO Stack up

Cold-plate with quick disconnects  
Supports two paddle cards

32-channels per paddle cards  
Fits existing 4x8-channel SiPh

High-speed edge connector  
Separate power connector



# Top 10 Advantages of XPO

1. Four-fold increase in front panel and switch rack level density.
2. Integrated cold plate which cools both low and high-power optics modules.
3. Supports Scale-Up, Scale-Out and Scale Across use cases.
4. Supports any kind of optics: DR, FR, LR, SR, ZR, CL and slow & wide
5. Supports copper cables and RF-Microwave.
6. Higher reliability due to lower temperatures and fewer components
7. Low-cost edge-connector paddle-card design
8. 50V supply voltage that eliminates motherboard voltage converters
9. Path to future 400G/lane modules that double capacity to 25.6 Tbps
10. Vibrant MSA Ecosystem

# XPO Supports all Optics Technologies

| Technolovgy        | Reach      | pJ/Bit | Failure Rate | Cost       |
|--------------------|------------|--------|--------------|------------|
| Passive Copper     | 2 m        | 0      | Lowest       | \$         |
| Active Copper      | 5 m        | 3-4    | Very Low     | \$\$       |
| RF-Microwave       | 10-20 m    | 3-4    | Very Low     | \$\$       |
| VCSEL              | 20-30 m    | 3-4    | Low          | \$\$       |
| DR8-LPO            | 2km        | 4-6    | Low          | \$\$\$     |
| FR4/LR4-LPO        | 2km        | 4-6    | Low          | \$\$\$     |
| CL (Coherent Lite) | 10-20 km   | 15     | Higher       | \$\$\$\$   |
| ZR (Coherent)      | 20-1000 km | 20-30  | Higher       | \$\$\$\$\$ |

# 25.6T XPO with 400G/Lane

Performance looks good for 400G-PAM6 (85 GHz Nyquist)  
Supports 400G copper cables and fully retimed DSP optics,  
including Coherent, inverse Gearbox and other DSP optics  
Expect to have measured results end of September

IEEE Hot Interconnects Symposium 2026, Lighting Session, Aug 19, 2026

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# XPO Card-edge 448G Simulation Results--Differential Insertion Loss



- IL : 448G XPO Rev.A1 delivers the optimal performance.

# XPO Conclusions

1. Highest Density Pluggable Optics form factor in the Industry
2. 204.8T per 10U with 200G/lane, 409.6T per 10U with 400G/lane
3. Supports all existing and emerging optics and interconnect standards
4. Liquid cooling up to 500W per module (for 25.6T coherent-lite)
5. Much lower failure rates with fewer components and lower temperatures
6. Vibrant XPO MSA ecosystem supported by all optics module vendors

## XPO MSA Core Optics Module Members

**Accelink**

Adtran

Amphenol

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TECHNOLOGIES

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**ciena**

**COHERENT**

**eoptolink**<sup>®</sup>



**Genuine**  
optics

**LIGENT**

**LIGHTMATTER**



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# Thank You!

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